Nonexistence of consecutive powerful triplets around cubes with mixed prime factorizations

Berkay Yüksel Sayim
Independent Researcher, Germany
berksa@tutamail.com · ORCID 0009-0004-4993-7352

June 2026

Abstract

A positive integer is *powerful* if every prime in its factorization occurs with exponent at least two. The Erdős–Mollin–Walsh conjecture asserts that no three consecutive integers are all powerful. Chan (2025) excluded triples of the form $x^3-1 = p^3y^2,\ x^3,\ x^3+1 = q^3z^2$, and She (2025) excluded $x^3 - 1 = p^2a^3,\ x^3,\ x^3 + 1 = q^2b^3$, with p, q prime. We close the remaining *mixed* configurations: there are no consecutive powerful triples of the form $x^3-1 = p^2a^3,\ x^3+1 = q^3z^2$, nor of the form $x^3-1 = p^3y^2,\ x^3+1 = q^2b^3$. The case analysis reduces both families to the single Diophantine equation $t^6+t^3+1 = 3w^2$, which we solve completely: exploiting the palindromic symmetry of the genus-2 curve $3w^2 = t^6 + t^3 + 1$, we descend to an elliptic quotient of Mordell–Weil rank 0 (curve `1296.k1` in the LMFDB, Cremona label `1296f2`), so the only rational solutions are $(t, w) = (1, \pm 1)$. We further show that no finite congruence sieve can decide this equation, explaining the necessity of the elliptic argument. As a complementary result we refine Beckon's mod-36 constraint on the smallest member of a hypothetical triple to arbitrary prime-square moduli and quantify the density decay of admissible residue classes.

1 Introduction

A positive integer m is *powerful* if $p^2 \mid m$ for every prime $p \mid m$; equivalently $m = a^2b^3$ with $a, b \in \mathbb{Z}_{>0}$ [7]. Pairs of consecutive powerful numbers, such as $(8, 9)$ and $(288, 289)$, occur infinitely often (e.g. via the Pell equation $x^2 - 8y^2 = 1$). In contrast, Erdős [5] conjectured, and Mollin and Walsh [9, 10] investigated systematically, that *no three consecutive integers are all powerful*. The conjecture remains open; under the *abc*-conjecture at most finitely many triples exist.

If $(n, n+1, n+2)$ is such a triple, the middle member may or may not be a perfect cube. The case in which the middle member is a cube x^3 has attracted recent attention. Chan [4] proved that there is no triple of the shape

$$x^3 - 1 = p^3y^2, \qquad x^3, \qquad x^3 + 1 = q^3z^2 \qquad (p, q \text{ prime}),$$

and She [12] proved the analogous statement for

$$x^3 - 1 = p^2a^3, \qquad x^3, \qquad x^3 + 1 = q^2b^3 \qquad (p, q \text{ prime}).$$

Both works treat *symmetric* configurations: the two outer members carry the same shape ($p^3 \cdot \square$ on both sides in [4]; $p^2 \cdot$ cube on both sides in [12]). Further recent activity in this circle includes van Doorn's study of three-term arithmetic progressions of consecutive powerful numbers [15]. The natural *mixed* configurations have, to our knowledge, not been treated. We close them.

Theorem 1. *There are no integers $x \geq 2$, primes p, q, and integers a, b, y, z such that either*

$$\text{(M1)} \qquad x^3 - 1 = p^2 a^3 \quad \text{and} \quad x^3 + 1 = q^3 z^2, \tag{1}$$

$$\text{(M2)} \qquad x^3 - 1 = p^3 y^2 \quad \text{and} \quad x^3 + 1 = q^2 b^3. \tag{2}$$

In particular, no triple of consecutive powerful numbers $(x^3 - 1,\, x^3,\, x^3 + 1)$ has mixed outer factorizations of these shapes.

The proof is a case analysis in the spirit of [4, 12] (Sections 3–4): factoring $x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ and tracking 3-adic valuations reduces all but one branch of each family to classical Mordell equations $Y^2 = X^3 + k$ with $k \in \{2, -2, 54, -54, -162\}$, whose integral points are completely known [6, 2, 8]. Both remaining branches lead to one and the same equation,

$$t^6 + t^3 + 1 = 3w^2, \tag{3}$$

with $t \geq 7$ in family (M1) and $t \leq -5$ in family (M2). Equation (3) concerns the ninth cyclotomic value $\Phi_9(t)$ and does not yield to the degree-4 or congruence techniques of [4, 12]; indeed we prove (Remark 14) that *no* finite congruence sieve can decide it. Instead we solve it completely:

Theorem 2. *The only solutions of $t^6 + t^3 + 1 = 3w^2$ in rational numbers are $(t, w) = (1, \pm 1)$.*

The proof (Section 5) exploits the palindromic symmetry $t \mapsto 1/t$ of the genus-2 curve $3w^2 = t^6 + t^3 + 1$: the curve admits two independent elliptic quotients over $\mathbb{Q}$, and one of them has Mordell–Weil rank 0 with trivial torsion — it is the curve `1296.k1` in the LMFDB [8], Cremona label `1296f2` — which pins down all rational points. Since $t = 1$ lies in neither branch range, both remaining branches are empty and Theorem 1 follows.

As a complementary result in the same circle of ideas we record a modular refinement of a little-known constraint of Beckon [1] on arbitrary consecutive powerful triples (not assuming a cube in the middle).

Proposition 3. *Let $(n, n+1, n+2)$ be powerful. Then, beyond Beckon's constraint $n \bmod 36 \in \{7, 27, 35\}$:*

(i) $n \bmod 900$ *lies in an explicit set of* 39 *residue classes, and* $n \bmod 44100$ *in an explicit set of* 1209 *classes;*

(ii) *for every prime $\ell \geq 5$, the proportion of admissible residues modulo ℓ^2 is $(\ell^2 - 3\ell + 3)/\ell^2$, while it is $1/4$ for $\ell = 2$ and $1/3$ for $\ell = 3$;*

(iii) *consequently the density of admissible classes modulo $\prod_{\ell \leq L} \ell^2$ tends to 0 as $L \to \infty$, but only at the rate $\asymp (\log L)^{-3}$; in particular no finite sieve of this type can resolve the Erdős–Mollin–Walsh conjecture.*

All computer calculations below are short and reproducible; the elliptic-curve computations were performed in SageMath and independently cross-checked (two independent runs, plus the LMFDB data [8]); the symbolic identities were verified exactly in SymPy. The verification scripts are provided as ancillary files.

2 Preliminaries

Throughout, $x \geq 2$ is an integer, p, q are primes, and v_ℓ denotes the ℓ-adic valuation. We collect standard facts; proofs are short and included for completeness where not cited.

Lemma 4 (parity). *In any triple of consecutive powerful numbers the even member is divisible by 4. If the middle member is x^3, then x is even; in particular $x^3 \pm 1$ are odd and coprime, so $p \neq q$ in (1)–(2).*

Proof. Exactly one or two of three consecutive integers are even; a powerful even number is $\equiv 0 \pmod 4$. If n and $n+2$ were both even, one of them would be $\equiv 2 \pmod 4$, hence not powerful. So the even member is unique, namely the middle one if the outer two are odd. For the middle member x^3: if x were odd, then $x^3 \pm 1$ would be even, contradiction. Finally $\gcd(x^3 - 1, x^3 + 1) \mid 2$ and both are odd. □

Lemma 5 (gcd and LTE). *$x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ with $g_\mp := \gcd(x \mp 1,\, x^2 \pm x + 1) = \gcd(x \mp 1, 3) \in \{1, 3\}$. Moreover, if $3 \mid x \mp 1$ then $v_3(x^2 \pm x + 1) = 1$ exactly.*

Proof. $x^2 \pm x + 1 = (x \mp 1)(x \pm 2) + 3$, giving the gcd claim (cf. [12, Lemma 4]). For the valuation: if $x = 1 + 3k$ then $x^2 + x + 1 = 3 + 9k + 9k^2 \equiv 3 \pmod 9$; if $x = -1 + 3k$ then $x^2 - x + 1 = 3 - 9k + 9k^2 \equiv 3 \pmod 9$. In both cases $v_3 = 1$ exactly. □

Lemma 6 (square sandwich). *For $x \geq 2$: $x^2 < x^2 + x + 1 < (x+1)^2$ and $(x-1)^2 < x^2 - x + 1 < x^2$. Hence neither $x^2 + x + 1$ nor $x^2 - x + 1$ is a perfect square.*

Lemma 7 (Chan-side structure). *Suppose $x^3 + 1 = q^3 z^2$ and $g_+ = 1$. Then $x + 1$ is a perfect square and $x^2 - x + 1 = q^{3+2k} \cdot \square$ for some $k \geq 0$. Analogously, if $x^3 - 1 = p^3 y^2$ and $g_- = 1$, then $x - 1$ is a perfect square and $x^2 + x + 1 = p^{3+2k} \cdot \square$.*

Proof. With $g_+ = 1$ the factors $x+1$ and $x^2 - x + 1$ are coprime, so each is, up to the prime q, a perfect square, and q^3 (odd exponent ≥ 3) divides exactly one of them. If q^3 divided $x + 1$ with $x^2 - x + 1$ a square, Lemma 6 is contradicted. The mirrored statement is identical. □

Lemma 8 (She's splitting; [12, Lemma 1]). *If $RS = p^2 C^3$ with $\gcd(R, S) = 1$, then one of R, S is a perfect cube and the other is p^2 times a perfect cube.*

Lemma 9 ([12, Lemma 2]). *$u^2 + u + 1 = 3v^3$ has only the integer solutions $u \in \{-2, 1\}$, and $u^2 - u + 1 = 3v^3$ only $u \in \{-1, 2\}$.*

Lemma 10 (Tzanakis; [14], cf. [12, Lemma 3]). *$u^2 + u + 1 = v^3$ has only $u \in \{-19, -1, 0, 18\}$, and $u^2 - u + 1 = v^3$ only $u \in \{-18, 0, 1, 19\}$.*

Lemma 11 (Mordell curves). *The integral points of $Y^2 = X^3 + k$ are: for $k = 2$: $(-1, \pm 1)$; for $k = -2$: $(3, \pm 5)$; for $k = 54$: $(3, \pm 9)$; for $k = -54$: $(7, \pm 17)$; for $k = -162$: none.*

Proof. All five values satisfy $|k| \leq 10^4$ and are covered by the complete solution of Mordell's equation in this range [6], extended to $|k| \leq 10^7$ in [2]; see also the corresponding LMFDB data [8]. (The case $k = -2$ is classical.) We re-verified all five lists with `integral_points` in SageMath in two independent runs. □

We will also use without further comment: squares mod 9 are $\{0, 1, 4, 7\}$; cubes mod 9 are $\{0, 1, 8\}$; cubes of integers $\equiv 1 \pmod 3$ are $\equiv \{1, 10, 19\} \pmod{27}$.

Exponent bookkeeping. For the Chan shape $x^3 \mp 1 = P^3Y^2$ one has $v_P \equiv 1 \pmod 2$, $v_P \geq 3$, and $v_r \equiv 0 \pmod 2$ for primes $r \neq P$; for the She shape $x^3 \mp 1 = P^2A^3$ one has $v_P \equiv 2 \pmod 3$ and $v_r \equiv 0 \pmod 3$ for $r \neq P$. Since $g_\mp \in \{1, 3\}$, every prime $r \neq 3$ divides exactly one of the two factors $x \mp 1$, $x^2 \pm x + 1$.

3 Family (M1): $x^3 - 1 = p^2a^3$, $x^3 + 1 = q^3z^2$

By Lemma 4, x is even and $p \neq q$. We split by x mod 3.

Case 1: $x \equiv 0 \pmod 3$ (so $g_- = g_+ = 1$)

By Lemma 7, $x + 1 = v^2$. By Lemma 8 applied to $(x - 1)(x^2 + x + 1) = p^2a^3$:

- (ii) $x - 1 = p^2u^3$, $x^2 + x + 1 = t^3$: Lemma 10 forces $x \in \{-19, -1, 0, 18\}$; only $x = 18$ is even and ≥ 2, but $x - 1 = 17$ is prime, not of the form p^2u^3. *Closed.*
- (i) $x - 1 = u^3$, $x^2 + x + 1 = p^2t^3$: combining with $x + 1 = v^2$ gives $v^2 - u^3 = 2$, a point on $Y^2 = X^3 + 2$; by Lemma 11 $u = -1$, $x = 0 < 2$. *Closed.*

Case 2: $x \equiv 1 \pmod 3$ (so $g_- = 3$, $g_+ = 1$)

Here $x^2 - x + 1 \equiv 1 \pmod 3$, so $q \neq 3$, and $x + 1 = v^2$ by Lemma 7; on the She side $v_3(x^2 + x + 1) = 1$ exactly (Lemma 5).

- $p = 3$: here $v_3(x^3 - 1) = 2 + 3v_3(a)$, while $v_3(x^3 - 1) = v_3(x - 1) + 1$ by Lemma 5; hence $v_3(x - 1) = 1 + 3v_3(a)$ and $v_3(x^2 + x + 1) = 1$. All prime exponents in $x^2 + x + 1$ other than that of 3 are $\equiv 0 \pmod 3$, so $x^2 + x + 1 = 3t^3$ with $3 \nmid t$; Lemma 9 gives $x \in \{-2, 1\}$, both < 2. *Closed.*
- $p \neq 3$, $p \mid x - 1$: then again $x^2 + x + 1 = 3t^3$ ($v_3 = 1$, $p \nmid$, all other exponents $\equiv 0$ mod 3), so $x \in \{-2, 1\}$. *Closed.*
- $p \neq 3$, $p \mid x^2 + x + 1$: then $v_3(x - 1) = 3v_3(a) - 1 \equiv 2 \pmod 3$ and all other prime exponents in $x - 1$ are $\equiv 0 \pmod 3$, hence $x - 1 = 9d^3$. But $x + 1 = v^2$ gives $v^2 = 9d^3 + 2 \equiv 2 \pmod 9$, impossible since squares mod 9 are $\{0, 1, 4, 7\}$. *Closed.*

Case 3: $x \equiv 2 \pmod 3$ **(so** $g_- = 1$**,** $g_+ = 3$**)**

She side (Lemma 8):

(ii) $x - 1 = p^2u^3$, $x^2 + x + 1 = t^3$: Lemma 10 gives $x \in \{-19, -1, 0, 18\}$; none is $\equiv 2 \pmod 3$, even, and ≥ 2. *Closed.*

(i) $x - 1 = u^3$ with u odd, $u \equiv 1 \pmod 3$ (so $u \equiv 1 \bmod 6$), and $x^2 + x + 1 = p^2t^3$. On the Chan side $g_+ = 3$ and $v_3(x^2 - x + 1) = 1$ exactly. Sub-split on q:

- $q = 3$: $x^3 + 1 = 27z^2$ gives $v_3(x + 1) = 2 + 2v_3(z)$ even and all other exponents in $x + 1$ even, so $x + 1 = m^2$; then $m^2 - u^3 = 2$, again $Y^2 = X^3 + 2$, so $u = -1$, $x = 0 < 2$. *Closed.*
- $q \neq 3$, $q^3 \mid x^2-x+1$: then $v_3(x+1)$ is odd, say $x+1 = 3^{2j+1}m^2$ with $\gcd(m, 3) = 1$ up to squares. If $j \geq 1$ then $27 \mid x + 1 = u^3 + 2$, i.e. $u^3 \equiv -2 \equiv 25 \pmod{27}$, impossible for $u \equiv 1 \pmod 3$. So $j = 0$, $x+1 = 3m^2$, i.e. $u^3 = 3m^2-2$; multiplying by 27: $(9m)^2 = (3u)^3 + 54$. By Lemma 11 the only points have $X = 3u = 3$, so $u = 1$, $x = 2$; but $x^3 - 1 = 7$ is not powerful. *Closed.*
- $q \neq 3$, $q^3 \mid x + 1$: then $x^2 - x + 1 = 3w^2$ ($v_3 = 1$ exactly, $q \nmid$, all other exponents even). With $x = u^3 + 1$:

$$(u^3 + 1)^2 - (u^3 + 1) + 1 = u^6 + u^3 + 1 = 3w^2,$$

  i.e. equation (3) with $t = u$. Here $u \geq 1$ since $x \geq 2$, and $u = 1$ gives $x = 2$, where $x^3 - 1 = 7$ is not of the form p^2a^3; together with $u \equiv 1 \pmod 6$ this forces $t = u \geq 7$. *This is the only surviving branch of (M1); it is empty by Theorem 2.*

4 Family (M2): $x^3 - 1 = p^3y^2$, $x^3 + 1 = q^2b^3$

Again x is even, $p \neq q$; split by x mod 3.

Case 1: $x \equiv 0 \pmod 3$ **(**$g_- = g_+ = 1$**)**

Chan side (Lemma 7): $x - 1 = u^2$ with u odd. She side (Lemma 8) applied to $(x + 1)(x^2 - x + 1) = q^2b^3$:

(b) $x + 1 = q^2s^3$, $x^2 - x + 1 = t^3$: Lemma 10 gives $x \in \{-18, 0, 1, 19\}$; no admissible value. *Closed.*

(a) $x + 1 = s^3$, $x^2 - x + 1 = q^2t^3$: then $s^3 - u^2 = 2$, i.e. a point on $Y^2 = X^3 - 2$; by Lemma 11 $(s, u) = (3, \pm 5)$, $x = 26 \equiv 2 \pmod 3$, contradicting the case. *Closed.*

Case 2: $x \equiv 1 \pmod 3$ **(**$g_- = 3$**,** $g_+ = 1$**)**

She side has $g_+ = 1$:

(b) $x + 1 = q^2s^3$, $x^2 - x + 1 = t^3$: as above, no admissible x. *Closed.*

(a) $x+1 = s^3$ with s odd, $s \equiv 2 \pmod 3$, hence $s \equiv 5 \pmod 6$ and $s \geq 5$; and $x^2-x+1 = q^2t^3$. On the Chan side $v_3(x^2+x+1) = 1$ exactly. Sub-split on p:

- $p = 3$: then $v_3(x-1)$ is even and all other exponents in $x-1$ are even, so $x-1 = u^2$; then $u^2 = s^3-2$, so $(s,u) = (3, \pm 5)$ by Lemma 11, giving $x = 26 \equiv 2 \pmod 3 \neq 1$. *Closed.*
- $p \neq 3$, $p \mid x-1$: then $x^2+x+1 = 3w^2$ ($v_3 = 1$, $p \nmid$, rest even). With $x = s^3-1$:

$$(s^3-1)^2 + (s^3-1) + 1 = s^6 - s^3 + 1 = 3w^2,$$

i.e. equation (3) with $t = -s \leq -5$. *This is the only surviving branch of (M2); it is empty by Theorem 2.*
- $p \neq 3$, $p \mid x^2+x+1$: then $v_3(x-1)$ is odd, $x-1 = 3^{2j+1}m^2$. If $j \geq 1$ then $27 \mid s^3-2$, so $s^3 \equiv 2 \pmod 9$, impossible (cubes mod 9). So $j = 0$, $s^3 = 3m^2+2$; multiplying by 27: $(9m)^2 = (3s)^3 - 54$ with $3 \mid X = 3s$. By Lemma 11 the only integral points have $X = 7$, not divisible by 3. *Closed.*

Case 3: $x \equiv 2 \pmod 3$ ($g_- = 1$, $g_+ = 3$)

Chan side: $x-1 = u^2$ (u odd), $x^2+x+1 = p^{3+2k} \cdot \square$. She side: $v_3(x^2-x+1) = 1$ exactly. We first dispose of the case $q = 3$:

- $q = 3$: here $x^3+1 = 9b^3$, so $v_3(x^3+1) = 2 + 3v_3(b)$, while $v_3(x^3+1) = v_3(x+1)+1$ by Lemma 5; hence $v_3(x+1) = 1 + 3v_3(b)$ and $v_3(x^2-x+1) = 1$, with all other prime exponents in x^2-x+1 being $\equiv 0 \pmod 3$. Thus $x^2-x+1 = 3s^3$, and Lemma 9 gives $x \in \{-1, 2\}$; $x = 2$ fails since $x^3-1 = 7$ is not of the form p^3y^2. *Closed.*

- $q \neq 3$: from $x^3+1 = q^2b^3$ and $v_3(x^2-x+1) = 1$ we get $v_3(x+1) = 3v_3(b) - 1 \equiv 2 \pmod 3$, in particular $9 \mid x+1$. Sub-split:
  - $q \mid x+1$: then $x^2-x+1 = 3t^3$ ($v_3 = 1$, $q \nmid$, all other exponents $\equiv 0 \bmod 3$), so $x \in \{-1, 2\}$ by Lemma 9; as before. *Closed.*
  - $q \mid x^2-x+1$: then $x+1 = 3^{3v_3(b)-1} \cdot s^3 = 9c^3$ for an integer c, and $x-1 = u^2$ gives $u^2+2 = 9c^3$; multiplying by 81: $(9u)^2 = (9c)^3 - 162$. By Lemma 11, $Y^2 = X^3 - 162$ has no integral points at all. *Closed.*

This completes the case analysis: both families reduce to the branches recorded above, all of which are empty except the two leading to (3).

5 The equation $t^6 + t^3 + 1 = 3w^2$

Write $\Phi_9(t) = t^6 + t^3 + 1$ for the ninth cyclotomic polynomial, so (3) reads $\Phi_9(t) = 3w^2$. The smooth projective model of

$$\mathcal{C} : \; Y^2 = 3t^6 + 3t^3 + 3 \qquad (Y = 3w)$$

is a curve of genus 2. Since the right-hand side is palindromic, $\mathcal{C}$ carries the involution $\sigma : (t, Y) \mapsto (1/t, \, Y/t^3)$ defined over $\mathbb{Q}$, and hence (besides the hyperelliptic involution ι) the second involution $\tau = \sigma\iota$. The quotients $\mathcal{C}/\sigma$ and $\mathcal{C}/\tau$ are elliptic, and $\mathrm{Jac}(\mathcal{C})$ is isogenous over $\mathbb{Q}$ to their product. Concretely:

Lemma 12. *Set $u = t + 1/t$, and*

$$V = \frac{Y(t+1)}{t^2}, \qquad W = \frac{Y(t-1)}{t^2}.$$

Then on $\mathcal{C}$ one has the exact identities

$$V^2 = 3(u+2)(u^3 - 3u + 1) =: q_+(u), \qquad W^2 = 3(u-2)(u^3 - 3u + 1) =: q_-(u).$$

Proof. Multiply $Y^2 = 3(t^6 + t^3 + 1)$ by $(t \pm 1)^2/t^4$ and use

$$(t \pm 1)^2 \, (t^6 + t^3 + 1) = t^4 \, (u \pm 2) \, (u^3 - 3u + 1),$$

which is a polynomial identity in t after substituting $u = t + 1/t$ (verified exactly; it follows from $u^3 - 3u + 2 = (u-1)^2(u+2)$ and $u^3 - 3u - 2 = (u+1)^2(u-2)$ together with $t^3 + t^{-3} = u^3 - 3u$ and $(t \pm 1)^2/t = u \pm 2$). $\square$

Proof of Theorem 2. Let $(t, w) \in \mathbb{Q}^2$ satisfy (3) and put $Y = 3w$. Note $t \neq 0$ (else $3w^2 = 1$) and $u = t + 1/t \neq -1$ (else $t^2 + t + 1 = 0$, no rational root). By Lemma 12, (u, W) is a rational point on the genus-one quartic curve

$$\mathcal{D}_- : \; W^2 = q_-(u) = 3(u-2)(u^3 - 3u + 1).$$

The quartic q_- is squarefree (its cubic factor has discriminant 81 and does not vanish at $u = 2$), so $\mathcal{D}_-$ is smooth of genus one; its two points at infinity satisfy $W_\infty^2 = 3$ and are conjugate over $\mathbb{Q}(\sqrt{3})$, hence not rational. $\mathcal{D}_-$ has the rational point $(2, 0)$.

We claim $\mathcal{D}_-(\mathbb{Q}) = \{(2, 0)\}$. Substituting $u = 2 + 1/T$, $W = S/T^2$ transforms $\mathcal{D}_-$ birationally into

$$S^2 = 9T^3 + 27T^2 + 18T + 3,$$

and the further substitution $(x, y) = (9T + 9, \, 9S)$ gives the elliptic curve

$$E : \; y^2 = x^3 - 81x + 243,$$

of conductor 1296. This is the curve `1296.k1` in the LMFDB [8] (Cremona label `1296f2`): it has Mordell–Weil rank 0 and trivial torsion, so $E(\mathbb{Q}) = \{\mathcal{O}\}$ — there is no affine rational point. (We verified rank 0 and triviality of torsion independently in SageMath with `rank(proof=True)`.) Running the birational maps backwards, any rational point of $\mathcal{D}_-$ with $u \neq 2$ would produce an affine rational point of E; hence $u = 2$ is the only possibility, and then $W^2 = q_-(2) = 0$.

Pulling back to $\mathcal{C}$: $u = t + 1/t = 2$ forces $(t-1)^2 = 0$, so $t = 1$ and $Y^2 = 9$, $w = \pm 1$. $\square$

Remark 13. The companion quotient $\mathcal{D}_+ : V^2 = q_+(u)$ has Jacobian $y^2 = x^3 - 189x + 999$ of rank 1 (LMFDB `1296.b1`, Cremona `1296l2`); this is why the descent must go through $\mathcal{D}_-$.

Remark 14 (no congruence sieve suffices). Equation (3) cannot be resolved by congruences alone, in the following precise sense. Substituting $z = t^3$, any solution gives a point on the conic $z^2 + z + 1 = 3w^2$; writing $2z + 1 = 3c$, $2w = b$ reduces the conic to the Pell equation $b^2 - 3c^2 = 1$, whose solutions with b even form a single orbit $\{z_k\}_{k \in \mathbb{Z}}$ of the recurrence $z_{k+1} = 14z_k - z_{k-1} + 6$ through $(z_0, z_1) = (-2, 1)$, the sign choice corresponding to the symmetry $z \mapsto -1-z$. Asking whether (3) has a solution with $t \neq 1$ is asking whether the orbit contains a cube other than $z_1 = 1$. Since $z_1 = 1$ *is* a cube, its residue class modulo any period of the sequence is a "shadow" that no congruence can eliminate — and via the symmetry it also protects a class in the negative branch. Hence no finite system of congruence conditions can prove the nonexistence of further cubes in the orbit, and some global (here: elliptic) input is unavoidable. A direct check of the recurrence (in exact integer arithmetic) confirms that no z_k with $k \leq 2000$ other than z_1 is a cube, the largest candidates having more than 2200 decimal digits.

Remark 15 (relation to Nagell–Ljunggren and to powers in recurrences). Since $\Phi_9(t) = (t^9 - 1)/(t^3-1)$, equation (3) is adjacent to the classical equations $(x^n-1)/(x-1) = 3y^m$ of Nagell–Ljunggren type. Note however that Nagell's theorem on $x^2+x+1 = 3y^n$ $(n > 2)$ does not apply: in (3) the square sits on the right-hand side ($m = 2$), while the cube condition sits inside the conic variable $z = t^3$. In the language of Remark 14, (3) asks for the cubes in a fixed binary recurrence orbit; the effective methods underlying the finiteness results of Pethő [11] and Shorey–Stewart [13] guarantee that there are only finitely many such cubes, but do not by themselves produce the explicit solution set. For the closely related factorisation $(2z+1)(z^2+z+1) = y^\ell$ see Bennett, Patel, and Siksek [3]. We are not aware of a published complete determination of the integer (or rational) solutions of (3); the contribution here is the explicit resolution via the rank-0 elliptic quotient.

6 A modular refinement of Beckon's condition

Beckon [1] observed that the smallest member n of a triple of consecutive powerful numbers must satisfy $n \equiv 7, 27, 35 \pmod{36}$; this elementary but useful constraint appears not to have been taken up in the recent literature. The underlying mechanism generalizes to any prime-square modulus.

Proof of Proposition 3. If m is powerful and ℓ prime, then $v_\ell(m) \neq 1$; in particular, if $m \equiv c \pmod{\ell^2}$ with $\ell \mid c$, $\ell^2 \nmid c$, then m is not powerful. Call $r \bmod \ell^2$ *admissible* if none of $r, r{+}1, r{+}2$ is $\equiv \ell c' \pmod{\ell^2}$ with $\ell \nmid c'$. For $\ell \geq 5$ the $\ell{-}1$ forbidden residues $\ell c'$ are pairwise at distance ≥ 5, so the $3(\ell{-}1)$ shifted prohibitions are distinct, leaving $\ell^2 - 3(\ell-1) = \ell^2 - 3\ell + 3$ admissible classes. For $\ell = 2$: n must be odd and $n{+}1 \equiv 0 \pmod 4$, leaving the class $3 \bmod 4$ (1 of 4). For $\ell = 3$: the windows avoiding $\{3, 6\} \bmod 9$ start at $r \in \{0, 7, 8\}$ (3 of 9). By CRT the admissible classes multiply: modulo 36 one obtains exactly Beckon's $\{7, 27, 35\}$; modulo $900 = 2^2 3^2 5^2$ there are $3 \cdot 13 = 39$ classes; modulo $44100 = 2^2 3^2 5^2 7^2$ there are $39 \cdot 31 = 1209$. (The explicit lists were computed by two independent implementations with matching output.) Finally, $\prod_{5 \leq \ell \leq L}\left(1 - \frac{3}{\ell} + \frac{3}{\ell^2}\right) \asymp (\log L)^{-3}$ by Mertens' theorem, which proves (iii). $\square$

7 Concluding remarks

Together with [4] and [12], Theorem 1 completes the four shape-combinations $\{p^3 \cdot \square, p^2 \cdot \text{cube}\}^2$ for consecutive powerful triples around a cube in which both outer members are divisible by a single controlling prime to an exact pattern. Natural next steps in this program include the relaxations $p = q$ and composite cofactors, and She's question whether $x^n - 1$ can be powerful for $x > 1$, $n > 2$. The bielliptic reduction of Section 5 may be of independent use for other palindromic sextic obstructions in this circle of problems.

Acknowledgements. Substantial parts of this work, including the reduction of Section 5, were developed with the assistance of Claude (Fable 5 model, Anthropic). All steps were independently verified as described above: SageMath (`integral_points`, `rank` with `proof=True`, in two independent runs), exact symbolic identities in SymPy, and cross-checks against the LMFDB. The author reviewed all arguments and takes full responsibility for the content.

References

[1] E. Beckon, *On consecutive triples of powerful numbers*, Rose–Hulman Undergrad. Math. J. **20** (2019), no. 2, Article 3.

[2] M. A. Bennett and A. Ghadermarzi, *Mordell's equation: a classical approach*, LMS J. Comput. Math. **18** (2015), 633–646.

[3] M. A. Bennett, V. Patel, and S. Siksek, *Perfect powers that are sums of consecutive cubes*, Mathematika **63** (2017), no. 1, 230–249; arXiv:1603.08901.

[4] T. H. Chan, *A note on three consecutive powerful numbers*, Integers **25** (2025), #A7; arXiv:2503.21485.

[5] P. Erdős, *Problems and results on consecutive integers*, Publ. Math. Debrecen **23** (1976), no. 3–4, 271–282.

[6] J. Gebel, A. Pethő, and H. G. Zimmer, *On Mordell's equation*, Compositio Math. **110** (1998), 335–367.

[7] S. W. Golomb, *Powerful numbers*, Amer. Math. Monthly **77** (1970), 848–852.

[8] The LMFDB Collaboration, *The L-functions and modular forms database*, `https://www.lmfdb.org`, 2026. Elliptic curves `1296.k1` and `1296.b1`.

[9] R. A. Mollin and P. G. Walsh, *On powerful numbers*, Internat. J. Math. Math. Sci. **9** (1986), 801–806.

[10] R. A. Mollin and P. G. Walsh, *A note on powerful numbers, quadratic fields and the Pellian*, C. R. Math. Rep. Acad. Sci. Canada **8** (1986), 109–114.

[11] A. Pethő, *Perfect powers in second order linear recurrences*, J. Number Theory **15** (1982), no. 1, 5–13.

[12] J. She, *Nonexistence of consecutive powerful triplets around cubes with prime-square factors*, Integers **25** (2025), #A103; arXiv:2507.16828v3.

[13] T. N. Shorey and C. L. Stewart, *On the Diophantine equation $ax^{2t} + bx^ty + cy^2 = d$ and pure powers in recurrence sequences*, Math. Scand. **52** (1983), no. 1, 24–36.

[14] N. Tzanakis, *The diophantine equation $x^3 - 3xy^2 - y^3 = 1$ and related equations*, J. Number Theory **18** (1984), no. 2, 192–205.

[15] W. van Doorn, *Three-term arithmetic progressions of consecutive powerful numbers*, arXiv:2605.06697 (2026).